

FLORENCE BOCKTING

Postdoctoral Researcher & Research Software Engineer

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Profile

Research scientist and research software engineer specialising in Bayesian methods, computational modelling, and reproducible research software. Currently a postdoc and research software engineer in the Bayesian Workflow Group at Aalto University, implementing Bayesian methods in open-source, well-tested software within the Stan ecosystem. Doctorate in computational statistics from TU Dortmund University, with research on simulation-based expert prior elicitation and authorship of the Python package `elicito`, distributed via PyPI and conda-forge. Background spanning psychology, cognitive science, and climate science, with experience contributing to open-source projects in Python, R, and C++.

Core Expertise

- Bayesian statistics
- Computational modelling
- Machine learning
- Research software engineering
- Open-source scientific software
- Reproducible and open science
- Data pipelines
- Cognitive, social, and climate science

Programming & Tools

Languages	Python, R, Stan; basic knowledge of C++ and OCaml
Version control & CI	Git, GitHub, GitLab, GitHub Actions
Testing & code quality	`pytest`, `testthat`, `ruff`, `black`, `mypy`, `lintr`
Packaging & environments	`uv`, `pip`, `poetry`, `pixi`, `renv`, PyPI, conda-forge, CRAN
Documentation	Sphinx, Read the Docs, `roxygen2`, R vignettes
Writing & reporting	Quarto, Jupyter, Jupytext, R Markdown, LaTeX, Overleaf, Shiny
Pipelines & compute	Prefect, `targets`, HPC (LIDO3, NERSC), Slurm clusters
Open science & data	Zenodo, OSF
Environments	Linux / Ubuntu, Windows
Lifecycle management	Reverse dependency checks, migration guides, deprecation management

Further Skills

Languages	German (native), English (fluent), French (intermediate)
Project management	Jira, Confluence, Notion
Communication	Slack, Zulip, Zoom
Writing & references	Zotero, Microsoft Office, LibreOffice / OpenOffice
Graphics	GIMP

Experience

Postdoctoral Researcher / Research Software Engineer 2026–present
Bayesian Workflow Group, Department of Computer Science, Aalto University
Espoo, Finland · PI: Prof. Dr. Aki Vehtari

- Implement Bayesian statistical methods as tested, documented, open-source software, primarily within the Stan ecosystem.
- Develop and maintain R packages by translating current Bayesian workflow research into robust software and improving existing package architecture, tests, and documentation.

Research Scientist / Research Software Engineer 2025–2026
Climate Resource GmbH
Berlin, Germany

- Developed greenhouse gas concentration data pipelines for earth-system modelling groups.
- Designed and implemented an ESA-funded pipeline for scraping, preprocessing, and assimilating multi-source observational data from ground-based and satellite sources, resulting in a unified, standardised data product.

Consultant Aug–Dec 2025
Jacobs Foundation
Zürich, Switzerland

- Implemented expert prior elicitation to support internal decision-making processes.

Research Scientist — 2023–2025

Computational Statistics, Department of Statistics, TU Dortmund University
Dortmund, Germany · PI: Prof. Dr. Paul-Christian Bürkner

SC SimTech, University of Stuttgart — 2022–2023

Stuttgart, Germany · PI: Dr. Paul-Christian Bürkner

- Conducted doctoral research in computational statistics on simulation-based expert prior elicitation.
- Developed a method for learning prior distributions in Bayesian models from expert knowledge.
- Implemented the approach as the open-source Python package `elicito`, distributed via PyPI and conda-forge.

Research Scientist — 2020–2022

Methods & Statistics, Department of Psychology, Philipps University of Marburg
Marburg, Germany · PI: Prof. Dr. Daniel W. Heck

- Worked on statistical method development, computational and mathematical modelling, social psychology, and the formalisation of verbal theories.
- Formalised a verbal theory of truth judgments and the truth effect in social psychology by developing a computational model in R.

Tutor & Student Assistant — 2015–2020

Harz University of Applied Sciences; University of Osnabrück
Wernigerode and Osnabrück, Germany

- Supported lecture preparation and led exercise seminars across courses including Data Ethics, Statistics, Computational Data Analysis, Bayesian Data Analysis, General Psychology, and Experimental Psychology.

Research Assistant — 2018–2019

Quantitative and Qualitative Healthcare, Produkt+Markt GmbH
Osnabrück, Germany

- Conducted qualitative and quantitative market research in the healthcare sector.
- Coded qualitative questionnaires, performed descriptive data analyses, and prepared presentations and reports.

Project Member — Student Skills Matching Platform — 2018

Harz University of Applied Sciences
Wernigerode, Germany

- Contributed to the development of a platform for matching students based on skills.
- Carried out a needs analysis and led the conceptual design of the matching system.

Internships

- **Qualitative Market Research in Healthcare**, Ipsos GmbH, Hamburg, Germany, 2017
- **General Psychology and Methodology**, University of Bamberg, Germany, PI: Prof. Dr. Claus-Christian Carbon, 2016

Selected Software and Open-Source Contributions

Maintainer / Author

- **`elicito`** (Python) — Expert prior elicitation: learning prior distributions of model parameters from expert knowledge. Distributed via PyPI and conda-forge, documented on Read the Docs.
- **`gradient-aware-harmonisation`** (Python) — Harmonisation of two time series using zero- and first-order gradient information.
- **`GHG-forcing-for-CMIP`** (Python) — Pipeline constructing a composite greenhouse gas dataset from ground-based and satellite observations, suitable as forcing input for CMIP climate models.

Contributor

- **Stan ecosystem** — `loo` (leave-one-out cross-validation), `bayesplot` (posterior and MCMC visualisation), `posterior` (draws manipulation), `brms` (Bayesian regression interface), and `stan-dev/math` (C++ autodiff and distributions): feature development, refactoring, testing, documentation, reverse dependency checks, and migration guides.
- **CMIP7 climate data** — `cmip7-scenariomip-ghg-concentrations` (ScenarioMIP forcing generation), `CMIP7-GHG-Concentration-Manuscripts` (reproducible Jupytext-to-LaTeX manuscript workflow), and `input4MIPs_CVs` (controlled vocabularies for input4MIPs metadata).
- **RSE survey analysis tooling** — Analysis of results from the International RSE Survey for the Nordic RSE community and development of a reproducible analysis tool for reuse across RSE communities.
- **Open-source community work** — Vignettes, tutorials, pull request reviews, issue triage, and mentoring.

Education

Doctorate in Computational Statistics — 2026
TU Dortmund University
Dortmund, Germany
Thesis *Simulation-based expert prior elicitation: Method and software development*
- Doctoral thesis conducted at SimTech / University of Stuttgart and TU Dortmund University.
- Awarded *magna cum laude*.

M.Sc. in Cognitive Science — 2020
University of Osnabrück
Osnabrück, Germany
- Majors: Cognitive modelling and artificial intelligence.
- Graduated with distinction.

B.Sc. in Business Psychology — 2018
Harz University of Applied Sciences
Wernigerode, Germany
- Majors: Market research and consumer behaviour.
- Graduated with distinction.

Marketing Communications Specialist — 2014
Dresden-Informatik GmbH
Dresden, Germany
- Vocational training certified by IHK, Chamber of Industry and Commerce, Germany.

Teaching & Thesis Supervision

Master Seminar on Multilevel Modelling — 2023–2024
- Target audience: Master's students in Data Science, Statistics, and Econometrics.
- Introduced the theory and analysis of multilevel models using R from both Bayesian and frequentist perspectives, implemented with `lme4` and `brms`.
- Language of instruction: English.

Programming Course: Introduction to Python — 2023–2025
- Target audience: Bachelor's and Master's students in Data Science, Statistics, and Econometrics.
- Covered Python fundamentals, data wrangling and visualisation, documentation with Sphinx, testing with `pytest`, and version control with Git and GitHub; course material published as a public repository.
- Languages of instruction: English and German.

Supervision of Theses — 2020–2025
- B.Sc. thesis: *Analysis of different initialization approaches for hyperparameter optimization with mini-batch stochastic gradient descent: A simulation study* · TU Dortmund University
- M.Sc. thesis: *Sensitivity analysis and performance evaluation of varying upper thresholds for discrete likelihoods using the Softmax-Gumbel Trick* · TU Dortmund University
- B.Sc. thesis: *The influence of response scales on the knowledge gain of underlying cognitive mechanisms: The role of uncertainty and truth perception in the Truth Effect* · Philipps University of Marburg
- B.Sc. thesis: *Empirical test of core assumptions of the Referential Theory: Influence of repetition on perceived coherence* · Philipps University of Marburg
- B.Sc. thesis: *Identification and testing of relevant psychological factors on truth judgments and the truth effect according to the Referential Theory* · Philipps University of Marburg
- B.Sc. thesis: *Truth Effect: The role of the response scale in truth effect designs with short delay* · Philipps University of Marburg

Publications & Talks

- Bockting, F., Bronder, S., Vehtari, A., & Ward, B. *Contributing to Stan: A Developer's Guide to the Core, Interfaces, and First Contributions.* Tutorial at StanCon26, Uppsala, Sweden. — 2026
- Bockting, F. *Simulation-based expert prior elicitation: Method and software development.* Doctoral thesis, TU Dortmund University.
- Bockting, F. *Predictive Prior Elicitation: State of the Art & Current Challenges.* Contributed talk, Workshop on Bayesian Modelling, Lund University. — 2025
- Bockting, F., & Bürkner, P. C. *elicito: A Python Package for Expert Prior Elicitation.* arXiv preprint.
- Bockting, F., Radev, S. T., & Bürkner, P. C. *Expert-elicitation method for nonparametric joint priors using normalizing flows. Statistics and Computing*, 35(5), 132.
- Bockting, F., Radev, S. T., & Bürkner, P. C. *Normalizing Flows for Simulation-Based Expert Prior Elicitation.* Contributed talk, MathPsych. — 2024

- Bockting, F., Radev, S. T., & Bürkner, P. C. *Simulation-Based Prior Knowledge Elicitation for Parametric Bayesian Models.* Contributed talk, ISBA.
- Bockting, F., Radev, S. T., & Bürkner, P. C. *Simulation-Based Prior Knowledge Elicitation for Parametric Bayesian Models.* Invited talk, Bayes@Lund.
- Bockting, F., Radev, S. T., & Bürkner, P. C. *Simulation-based prior knowledge elicitation for parametric Bayesian models. Scientific Reports*, 14, 17330.
- Heck, D. W., & Bockting, F. *Benefits of Bayesian model averaging for mixed-effects modeling. Computational Brain & Behavior*, 6(1), 35–49. 2023
- van Doorn, J., ..., Bockting, F., & Aust, F. *Bayes factors for mixed models: A discussion. Computational Brain & Behavior*, 6(1), 1–13.
- Bockting, F., & Heck, D. W. *Measuring Individual Differences in the Truth Effect: A Formal Analysis.* Fast talk at MathPsych. 2021
- Stephan, A., ..., Bockting, F., ... *Nachwort.* In A. M. Turing, *Computing Machinery and Intelligence. Können Maschinen Denken?* Reclam, pp. 131–201.

Workshops & Courses

- Developing Secure Software, OpenSSF, Linux Foundation 2026
- High Performance R, CSC – IT Center for Science, Finland
- Copyright for Computer Programs & Software, TU Dortmund University 2024
- Research Software Engineering Summer School, Karlsruhe Institute of Technology, KIT
- Theory of Science, Prof. Dr. Zoglauer, University of Stuttgart 2023
- Foundations of Deep Learning for the Social Sciences, University of Tübingen
- The Statistics Wars and Their Casualties, online seminar series, Prof. Dr. Deborah Mayo, Prof. Dr. Roman Frigg, and Prof. Dr. Margherita Harris
- Summer School on Advanced Bayesian Data Analysis with Stan, Dr. Bruno Nicenboim, University of Potsdam 2022
- Interval Hypothesis Testing, Prof. Dr. Daniël Lakens, Eindhoven University of Technology
- Robust Cognitive Bayesian Analysis, Prof. Dr. Jeffrey N. Rouder, University of California
- Bayesian Evaluation of Informative Hypotheses, Prof. Dr. Herbert Hoijtink, Utrecht University
- Multinomial-Processing-Tree Modeling: Foundations and Recent Advances, Prof. Dr. Edgar Erdfelder and Prof. Dr. Daniel Heck, University of Mannheim 2021
- Single- vs. Dual-Process Theories, Prof. Dr. Mandy Hütter, University of Tübingen
- Introduction to Bayesian Statistics, Prof. Dr. Daniel Heck, Philipps University of Marburg

Conferences & Events

- StanCon — Stan Community, Uppsala
- MathPsych — Society for Mathematical Psychology, Tilburg
- ISBA — International Society for Bayesian Analysis, Venice
- DGP — German Psychological Society, Leipzig
- EGU — European Geosciences Union, Vienna
- Bayes@Lund, Lund

Engagement

- Analysed results from the International RSE Survey for the Nordic RSE community and developed a reproducible analysis tool for reuse across RSE communities worldwide.
- Contributed to the HiRSE Summary of AI for RSE with a video on “AI-assisted contributions in Open Source”.
- Carpentries lesson maintainer for “HPC Workflow Management with Snakemake”.
- Mentor in Google Summer of Code 2026 for “Parallelizing `loo` package”.
- Co-instructor for the StanCon 2026 tutorial “Contributing to Stan: A Developer’s Guide to the Core, Interfaces, and First Contributions”.
- Author of “Contributing to Stan”, onboarding material guiding new contributors through their first contribution to the Stan ecosystem.

Awards & Volunteering

- Fellow of the Studienstiftung des deutschen Volkes (German Academic Scholarship Foundation), 2015–2020.
- Volunteer at KANA Dortmund (soup kitchen), Dortmund, Germany.
- Volunteer at a residential home for children and adolescents with disabilities, Lebenshilfe e.V.